EXTENSION & ENRICHMENT WORKBOOK

SWIM & WATER *SAFETY*

Health & Physical Education · Year 7–8 · Victorian Curriculum F–10 Version 2.0

This workbook goes **deeper** on every topic. You will analyse real data, think critically about why drowning happens, evaluate evidence, and produce a sophisticated communication product. Expect to research, calculate, argue, and justify — not just describe.

STUDENT NAME ____________

CLASS ____________

TEACHER ____________

Extension Workbook — Expectations: Answers should go beyond description. *Analyse, evaluate, justify, and use evidence.* Where a question asks "why," give a reasoned explanation with supporting data, not just a one-word answer. Research questions require a source URL.

1 Statistical Analysis — Victorian Drowning Data

VC2HP8P10 · VC2HP8P09

ADVANCED CONTEXT

→ 54 fatal drownings in Victoria 2023–24 — highest in over a decade

→ Victoria's population ≈ 6.9 million → drowning rate = 54 ÷ 69 = 0.78 per 100,000

→ Australian national rate: approximately 1.2 per 100,000 (RLSSA National Report)

→ Non-fatal drownings: 60% above 10-year average — far more common than fatal

→ CALD community representation: ~40% of deaths — highest proportion ever recorded

→ Males: 46 of 54 deaths (85%) — consistent with global pattern of ~80%

→ All 54 deaths at unpatrolled locations — 0 deaths at patrolled locations

→ WHO global rate: ~320,000 drowning deaths per year; 90% in low/middle-income countries

Sources: LSV — lsv.com.au/research/victorian-drowning-reports/ · RLSSA — royallifesaving.com.au · WHO — who.int/news-room/fact-sheets/detail/drowning

Activity 1.1 — Calculations & Interpretation

Individual · ~15 min

Question 1.1a — Calculation

2 marks

Using the population data above, calculate: *what percentage of the 54 Victorian drowning deaths were male?* Show your working.

Working: 46 ÷ 54 × 100 = ________________________%

Now compare this to the global figure (80%). What does the difference suggest about Victorian male risk-taking near water?

Victorian % is _______ than the global figure. This suggests
____

____

Question 1.1b — Rate Comparison

3 marks

Victoria's drowning rate is 0.78 per 100,000 people. The Australian national average is 1.2 per 100,000. Calculate how many Victorian lives per year would be SAVED if Victoria's rate dropped to match the national average. Show working and interpret the result.

Current deaths: 54 · Target rate: 1.2 per 100,000 × 6.9 million = _______ expected deaths

Lives saved = 54 - _______ = _______

Interpretation:

____

____

Question 1.1c — CALD Data Analysis

3 marks

CALD communities represented approximately 40% of Victorian drowning deaths in 2023–24. Victoria's CALD population is approximately 28% of the total population. Using this information, explain whether CALD Victorians are over-represented, under-represented, or proportionally represented in drowning deaths. Show your reasoning.

CALD % of drowning deaths: 40% · CALD % of population: 28%

This means CALD Victorians are: over-represented / under-represented / proportional (circle one)

Explanation:

____

____

Suggest TWO specific reasons (not just "language barriers") why this over/under-representation exists:

1. ____

2. ____

Question 1.1d — Data Visualisation

3 marks

In the space below, create a **bar graph** comparing Victorian drowning deaths by location type. Research the approximate figures from the LSV report (lsv.com.au/research/victorian-drowning-reports/). Label axes, include a descriptive title, and write a 2-sentence analysis beneath.

📊 *Draw your bar graph here — or attach on a separate page*

Analysis:

____

____

Source URL for your data: ____

2 Beach Flag System — Analysis & Critical Evaluation

VC2HP8P10

Activity 2.1 — Flag System Evaluation ★ COLOUR

Individual · ~10 min

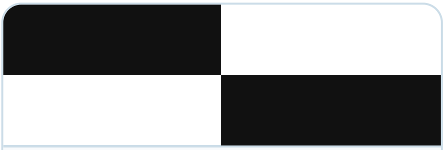
Beyond just identifying flags — analyse WHY each flag exists and WHO it protects. Complete the analysis column.

Red & Yellow

Why this flag exists + who it protects: ____________________

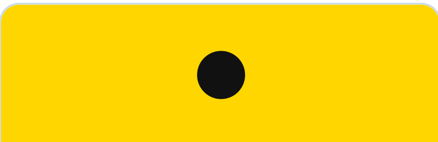
Single Red

Why this flag exists + limitation of this system: ______



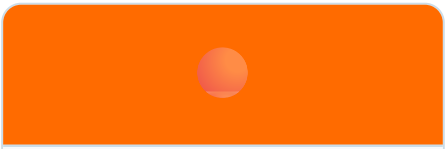
Black & White

Why separation of zones matters + risk if ignored: ______



Yellow + Black Ball

Who enforces these rules + what happens if no-one does: ____



Orange Windsock

Physics reason: why offshore wind + inflatable = drowning risk: ____

Purple Flag

Limitation: why is this flag less reliable than the others? ______

Question 2.1a — Critical Evaluation — 3 marks

The LSV data shows ALL 54 Victorian drowning deaths in 2023–24 occurred at unpatrolled locations. A student argues: "The flag system is pointless — people just ignore it anyway." Evaluate this argument. Do you agree or disagree? Use evidence.

I agree / disagree (circle) because:

____

Evidence:

____

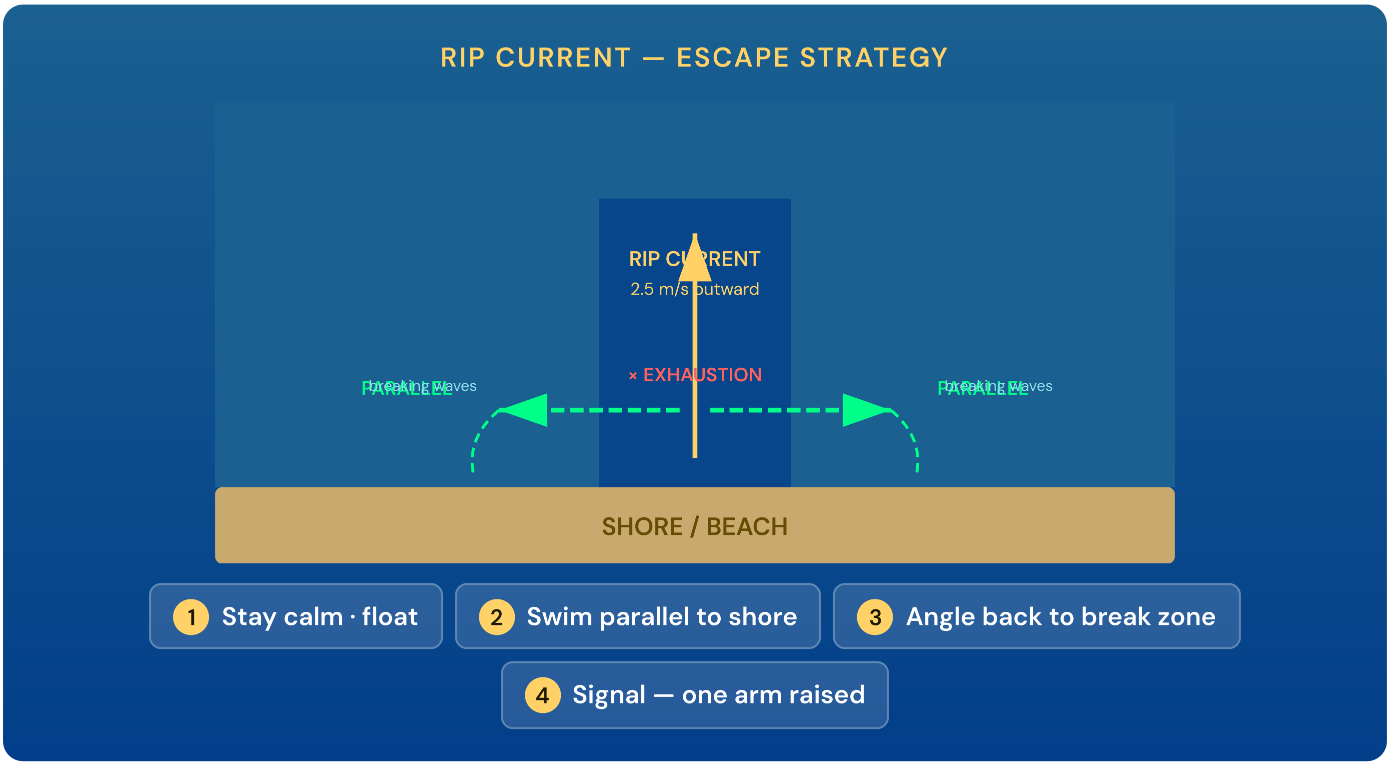
Counter-argument:

____

Activity 2.2 — Rip Current: Physics & Strategy ★ COLOUR

Individual · ~12 min

Rip currents can flow at up to 2.5 m/s. The world record 100m freestyle swim pace is approximately 2.0 m/s. Study the diagram and answer the analysis questions.



Question 2.2a — Physics Reasoning

3 marks

A rip current flows at 2.5 m/s. The world-record 100m swimmer averages approximately 2.0 m/s. Using this data, explain precisely why even a highly capable swimmer cannot escape a rip by swimming directly against it.

The rip current speed (2.5 m/s) is _______ than the world record swim speed (2.0 m/s).

This means that even the fastest swimmer in the world would move _______ m/s backward.

For an average Year 7 swimmer (~0.8 m/s), the net movement would be _______ m/s away from shore.

This leads to:

Question 2.2b — Strategy Evaluation

3 marks

The LSV advises floating and signalling for help rather than attempting to swim out of a rip if you are unable to escape in a few minutes. Evaluate this advice from a physiological standpoint — why does energy conservation matter more than attempting to escape?

📊 Research Extension

Visit the SLSA rip current page (sls.com.au/surf-safety/rip-currents/) and find ONE statistic about rip currents that is NOT in your workbook. Write it below with the exact URL.

New statistic:

Source URL:

3 Emergency Response — Advanced Application

VC2HP8P10 · VC2HP8M01

DEEPER KNOWLEDGE

→ **CPR physiology:** Chest compressions generate ~25–30% of normal cardiac output — enough to keep the brain alive until the heart restarts

→ **4-minute rule:** Brain cells begin dying after 4–6 minutes without oxygenated blood — bystander CPR is the single most important survival factor

→ **Bystander effect:** Research shows bystander CPR rates are higher when individuals feel personally responsible — diffusion of responsibility in a crowd reduces the chance anyone acts

→ **Compression rate:** 100–120 beats per minute; depth 5–6cm; allow full chest recoil between compressions

→ **AED:** Automated External Defibrillators will only deliver a shock if the heart is in a shockable rhythm — they are safe for untrained bystanders to use

Source: St John Ambulance Vic — stjohnvic.com.au · Heart Foundation — heartfoundation.org.au

Activity 3.1 — Advanced Emergency Analysis

Individual · ~15 min

Question 3.1a — Physiological Explanation

3 marks

Explain the physiological reason why *starting CPR within 4 minutes* is more important than starting CPR after 8 minutes, even if professional help arrives at both times. Use the terms: cardiac output, oxygenated blood, brain cells, irreversible damage.

Question 3.1b — Bystander Effect 3 marks

The "bystander effect" describes how people in a crowd are less likely to help in an emergency because they assume someone else will act. Explain how you would specifically overcome the bystander effect at a beach incident to ensure CPR is started quickly.

Strategy: Instead of calling out to "someone," I would

Why this works psychologically:

EXTENDED SCENARIO

A group of 6 students are at an unpatrolled beach. One student is pulled under by a wave near rocks and is dragged back to shore unconscious after 3 minutes. Several bystanders are watching. You have a phone. The nearest lifeguard is 2km away at a patrolled beach. The nearest AED is at the car park, 400m away.

Question 3.1c — Multi-Step Decision Making 5 marks

Describe the exact sequence of actions for the first 5 minutes. Assign roles to the 6 people present, justify your decisions, and explain what you would do differently if the victim was found not breathing.

Person	Assigned role	Why this role?
Person 1 (you)		
Person 2		
Person 3		
Persons 4–6		

If not breathing, the priority order changes because:

Question 3.1d — Throw Rescue Evaluation 3 marks

Rank and justify the throw rescue options at this scene. Available items: phone, rope attached to nearby boat, surfboard resting on beach, t-shirt, piece of driftwood. Explain the RLSSA hierarchy in your answer.

Item	Rank	Justification (with RLSSA principle)
Rope from boat		
Surfboard		
T-shirt		
Driftwood		
Phone		

4 Risk Analysis, Social Factors & Decision-Making

VC2HP8P08 · VC2HP8P10

Activity 4.1 — Social Determinants of Drowning Risk

Individual · ~15 min

Extension context: Drowning is not equally distributed across the population. Social factors — including gender norms, cultural background, and socioeconomic status — significantly influence who drowns and why. Research shows these factors often matter more than swimming ability alone.

Question 4.1a — Gender Analysis

3 marks

85% of Victorian drowning victims in 2023–24 were male. Research suggests this is partly explained by "masculinity norms" — social expectations about how males should behave. Explain what these norms are, how they increase drowning risk specifically, and propose ONE strategy a school could use to address this.

Question 4.1b — Hazard Mapping with Justification

4 marks

For each Victorian environment, identify THREE hazards, rank them H/M/L, and explain the mechanism by which each hazard causes drowning — not just that it's "dangerous."

Environment	Hazard	Rank	Mechanism (how it causes drowning)

River / Dam (e.g. Yarra, Eildon)			
Unpatrolled Beach (e.g. remote Victorian coast)			

Question 4.1c — Peer Pressure: Psychological Analysis 3 marks

Psychological research shows peer pressure is strongest when: (a) the group is watching, (b) there is a clear "dare" or challenge, and (c) refusing seems like weakness. Using this framework, design a TWO-STEP strategy for resisting a dare to swim at an unpatrolled location. Your strategy must address all three psychological triggers.

5 Inland Waterways — Physiology, Research & Rescue

VC2HP8P10 · VC2HP8M01

Activity 5.1 — Cold Water Shock: Physiological Analysis

Individual · ~15 min

Question 5.1a — Physiological Sequence 3 marks

Describe the complete physiological sequence that occurs when a person enters water at 12°C unexpectedly. Use the terms: gasping reflex, hyperventilation, peripheral vasoconstriction, cardiac stress, swimming failure. Explain why this is dangerous even in shallow water.

Question 5.1b — 1-10-1 Rule Application 2 marks

Explain the 1-10-1 cold water survival rule and describe how you would apply it if you fell into Lake Eildon in December. What decisions would you make at each stage?

Minute 1 (breathing control): ____

Minutes 1–10 (swimming window):

____

Hour 1 (hypothermia risk):

____

Question 5.1c — Research Investigation

3 marks

Visit Paddle Australia (paddle.org.au) and Parks Victoria (park.vic.gov.au). For ONE specific Victorian river or lake, research: (a) the specific hazards identified, (b) any seasonal factors (e.g. snowmelt, rainfall), and (c) any recent incidents if publicly available.

Waterway I researched:

____

Specific hazards:

____

Seasonal factors:

____

Relevant incident / safety note:

____

Source URLs:

____

Question 5.2 — Throw Rescue: Design Your Own Protocol

3 marks

You are kayaking on a remote Victorian river with a group of 4. One person's kayak overturns in cold water 15m from shore. They are conscious but struggling. Design a 3-step rescue protocol using the RLSSA hierarchy that accounts for the 15m distance, cold water, and remote location.

Reference: RLSSA — royallifesaving.com.au · Paddle Australia — paddle.org.au

Question 5.3 — Alcohol: Multi-Factor Risk Analysis

3 marks

Construct a multi-factor risk argument explaining why alcohol + cold Victorian river water is a particularly dangerous combination. Your answer must address at least FOUR distinct physiological or behavioural mechanisms and cite the relevant body system for each.

Activity 5.2 — Diving, Spinal Injury & Structural Risk Analysis

Individual · ~15 min

DATA CONTEXT — AUSTRALIAN DIVING & SPINAL INJURY

→ Unsafe diving accounts for approximately **10%** of all spinal cord injuries and **20%** of quadriplegia cases in Australia (AIHW Diving Injury Report)

→ **95%** of aquatic spinal injury patients are male; **75%+** are under 30; **66%** suffer permanent disability

→ Highest incidence age group: **children aged 10–14**

→ The cervical vertebrae (C4–C6) fracture or dislocate on head-first impact — compressing or severing the spinal cord. Spinal cord neurons cannot regenerate; the damage is permanent.

→ Three 2024–25 case studies: **Terrigal Wharf NSW** (wharf over shallow tidal water), **Torquay Front Beach VIC** (shifting surf sandbank), **Bawley Beach NSW** (sandbank formed between visits — false confidence from prior experience)

→ Core prevention message: **"Feet First, First Time, Every Time"** — and check depth on every visit, not just the first. Sandbanks, tidal variation, drought, flooding, and sediment change depth without warning.

Sources: AIHW aihw.gov.au · Royal Perth Hospital spinal injury review · RLSSA royallifesaving.com.au

Question 5.2a — Structural Analysis 3 marks

The data shows 95% of aquatic spinal injury patients are male and the highest-risk age group is 10–14 years. Using the social determinants framework, construct a structural explanation for this pattern. Your answer must go beyond "males take more risks" — identify the specific social factors that drive the behaviour and explain why individual education alone is insufficient as a prevention strategy.

Question 5.2b — False Confidence & Mechanism Analysis 3 marks

The Bawley Beach case (NSW 2024) illustrates *false confidence from prior experience*. Explain: (a) the specific physical mechanism by which a sandbank could form between visits and reduce water depth; (b) why prior safe experience at a location can be MORE dangerous than approaching it with no knowledge; and (c) what the "Feet First, First Time" message does and does not address as a prevention strategy.

Question 5.2c — Intersecting Risk Factors 4 marks

Using the data above and the social determinants content from Section 4, design a specific, evidence-based prevention strategy targeting the highest-risk group for aquatic spinal injury. Your strategy must: (a) name the target group precisely; (b) cite at least two statistics; (c) address both the physical hazard (depth change) and the social factor (masculinity norms/peer pressure); (d) include one SMART success metric; and (e) identify the most significant limitation of your approach.

6 Personal Action Plan & Communication Campaign

VC2HP8P09 · VC2HP8P10

🎯 Extension Assessment Task — Communication Campaign

Choose ONE option. All options require evidence-based argument, at least THREE credible sources with URLs, and a sophisticated understanding of your target audience.

Option A — Evidence-Based Social Campaign (targeted at a high-risk group)

Design a complete 3-slide social media campaign targeting ONE specific high-risk group identified in Victorian drowning data (e.g. young males, CALD communities, inland waterway users, alcohol-affected swimmers). Each slide must: address a specific barrier for that group, cite a statistic from LSV/WHO data, and include a measurable call to action. Justify your design choices in 100 words.

Option B — Policy Brief to a Government Minister

Write a 400–500 word policy brief to the Victorian Minister for Sport recommending TWO specific government actions to reduce the CALD drowning disproportionality. Your brief must: cite evidence, acknowledge counter-arguments, and propose a measurable success metric for each recommendation.

Option C — School Safety Audit Report ★

Conduct a water safety audit of your school: What does it currently provide? What are the gaps? Produce a structured 400-word report with specific recommendations benchmarked against Victorian Curriculum 2.0 requirements (VC2HP8P10, VC2HP8M01) and LSV classroom resources. Include a priority matrix.

Planning Space

Option chosen: _______ · Target audience / addressee: ______________________________

Central argument / key claim: ____

Source 1 (URL): ____

Source 2 (URL): ____

Source 3 (URL): ____

Evidence-based statistic I will use: ____

✏️ Extension Self-Assessment

Rate yourself honestly — this is for your own reflection.

I can calculate and interpret drowning rate statistics (per 100,000)	Developing	Solid	Can teach it
I can explain the physics of rip currents numerically	Developing	Solid	Can teach it
I can explain CPR physiology and the 4-minute rule	Developing	Solid	Can teach it
I can analyse social determinants of drowning risk	Developing	Solid	Can teach it
I can explain the 1-10-1 cold water rule and apply it	Developing	Solid	Can teach it
I can produce an evidence-based policy argument	Developing	Solid	Can teach it

One thing I still want to investigate: